START HERE - Python exam preparation pack

Everything is in `Practice/Exam_Prep_Pack/`. Printable PDFs are in `Exam_Prep_Pack/PDF_Print/` (one per document, A4, black & white friendly). Runnable, tested solutions are in `Practice/Solutions_py/`.

Situation in one paragraph

Your Python assessment next week is a **mock of the final lab exam**: one `.py` file, uploaded. **tkinter is no longer examined.** The newest paper (2024) is pandas -> matplotlib -> SQLite -> PyTorch, and the practice sheet "Task 3" shows PyTorch possibly being **replaced by CSV + pandas + numpy + matplotlib**. SQLite and "do the right thing when data is wrong" have appeared in every paper since 2021. So prepare for **both** variants of task 3.

The reading order

#	Document	Why
1	`00_Exam_Pattern_Analysis`	how the papers are built and marked; the 10 commandments; study plan
2	`Paper_Solutions/P1_2024_Paper_Explained`	the closest thing to your mock, every line explained with analogies
3	`Topics/T4_SQLite`	highest-yield topic
4	`Topics/T5_Pandas_NumPy`, `T6_Matplotlib`	Task 1 of any data paper
5	`Topics/T3_Files_CSV_Exceptions`	the Practice Task 3 solution + file handling
6	`Topics/T7_PyTorch_Basics`	the network task
7	`Mock_Exams/MockA`, then `MockB`	sit them timed, then self-mark
8	`Paper_Solutions/P2–P4` (2021-2023)	extra database/validation practice, console versions
9	`Topics/T1_Python_Core`, `T2_OOP_Python`	fill gaps; `T8_Tkinter_Legacy` only if you have spare time
10	`Exam_Day_Cheat_Sheet`	**print this; take it into the exam**

What to print (suggested, about 35 pages)

`Exam_Day_Cheat_Sheet` (3-4 pp), `P1` (~9 pp), `T4`, `T5`, `T6` (the rest as you wish). **Also hand-write the "Quick-reference box" from each topic onto your own notes sheet** - it is allowed and it is the best revision there is.

Folder map

```
Practice/
  START_HERE.md                   <- this file
  Solutions_py/                   <- all tested .py solutions
  Exam_Prep_Pack/
    00_Exam_Pattern_Analysis.md
    Exam_Day_Cheat_Sheet.md
    Paper_Solutions/  P0 index, P1 2024, P2 2021, P3 2022, P4 2023
    Topics/           T1 core, T2 OOP, T3 files/CSV, T4 SQLite, T5 pandas/numpy,
                      T6 matplotlib, T7 PyTorch, T8 tkinter (legacy)
    Mock_Exams/       README, MockA (2024-style), MockB (Practice-Task-3-style)
    PDF_Print/        printable PDFs of every .md above
    _tools/build_pdfs.py   rebuilds the PDFs (needs: pip install markdown pygments)
  (your original uploaded exam papers are untouched)
```

Exam environment checklist

- Know whether the exam machine has `pandas`, `matplotlib`, `torch` (open a terminal and `python -c "import pandas, matplotlib, torch"` in the first minute; if torch is missing, tell the invigilator).
- File name `your_name_python_task.py`; data files in the same folder as the script.

- Run the whole file from a clean start before uploading.

When you upload your own coursework code, tell me which parts and I will fold your own idioms into the topic documents.